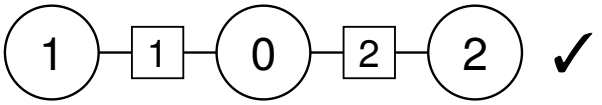
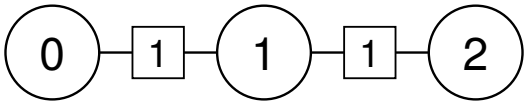
Put a different number card on each dot, choosing from the cards named under the network. In the box on each link, write the difference between the numbers at its two ends. The network is labelled when its link numbers are 1, 2, 3 and so on up to the number of links, each once.



✓

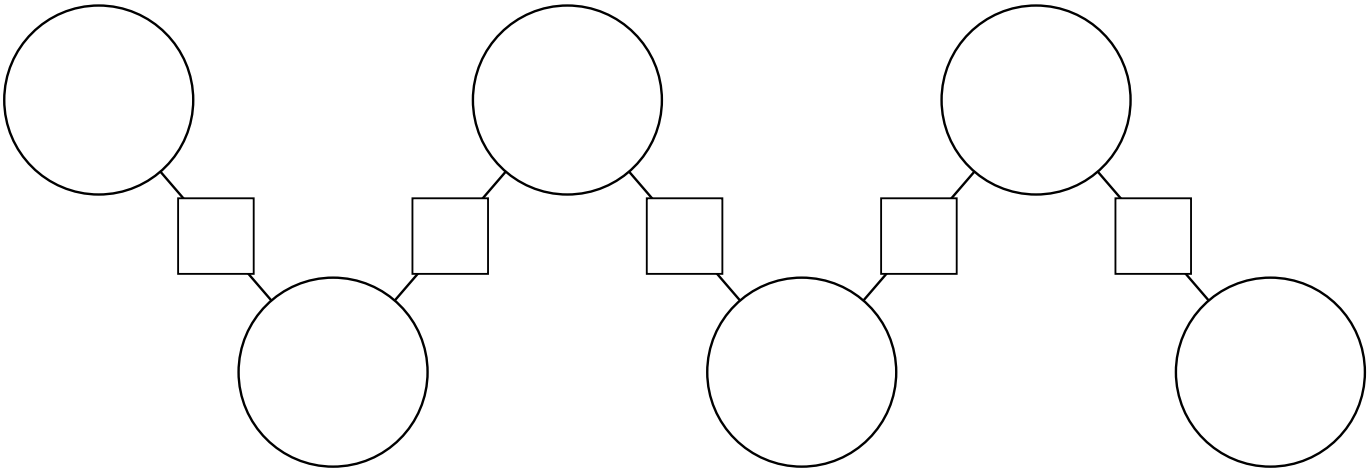
cards 0 to 2



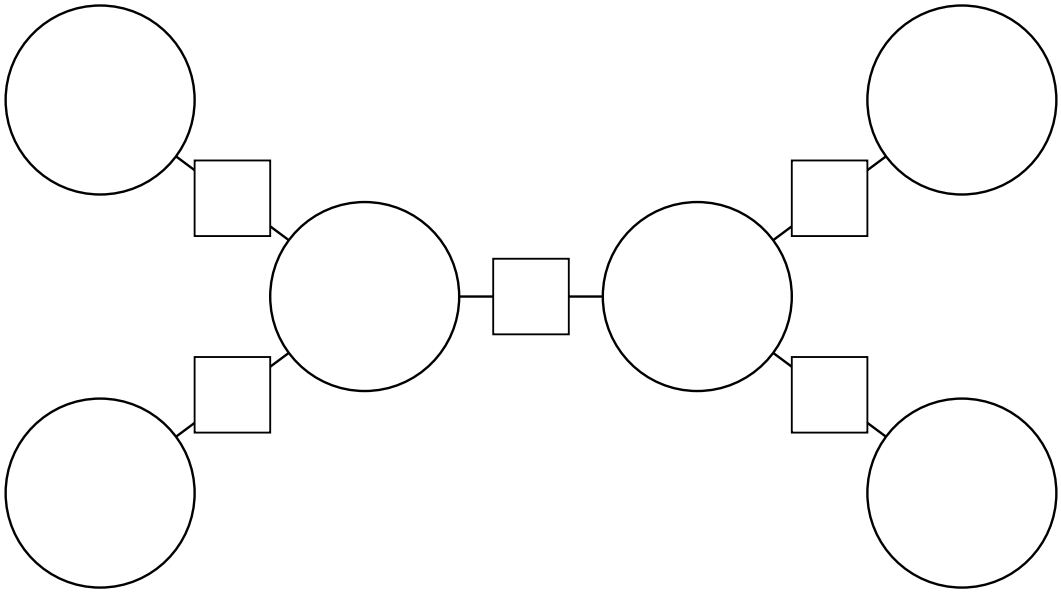
✗

cards 0 to 2

Problem 1: Label each network.

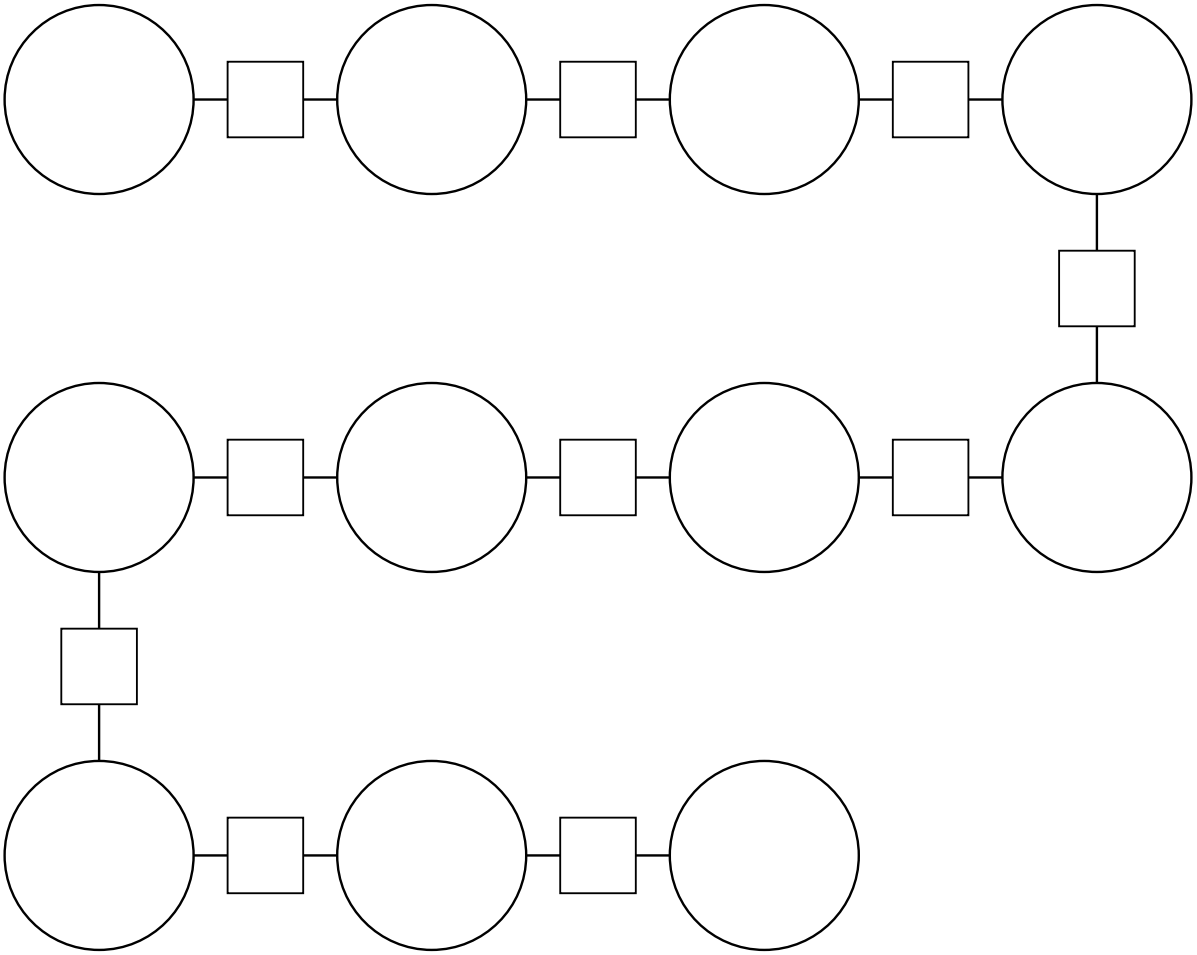


cards 0 to 5



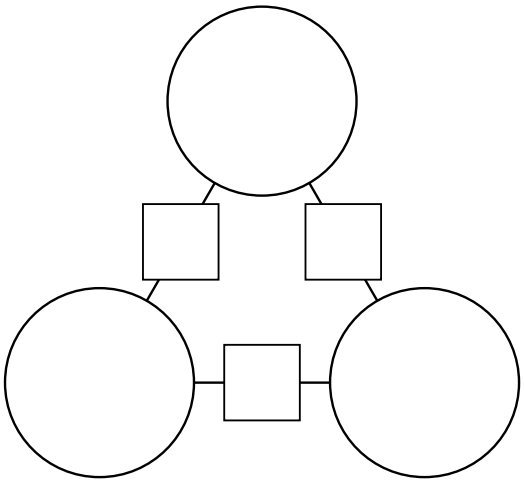
cards 0 to 5

Problem 2: Label this path. Then describe a way to label a path with any number of dots, and explain why it works.

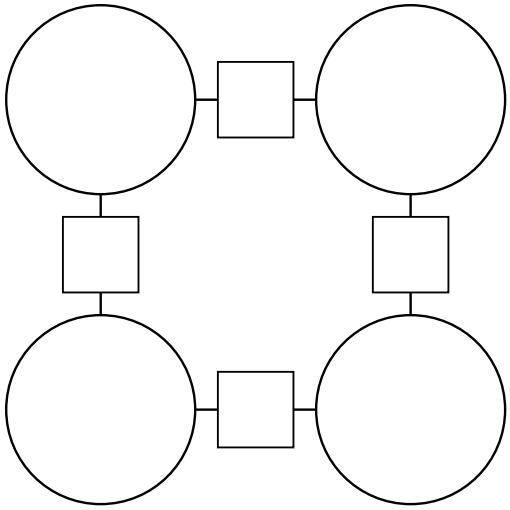


cards 0 to 10

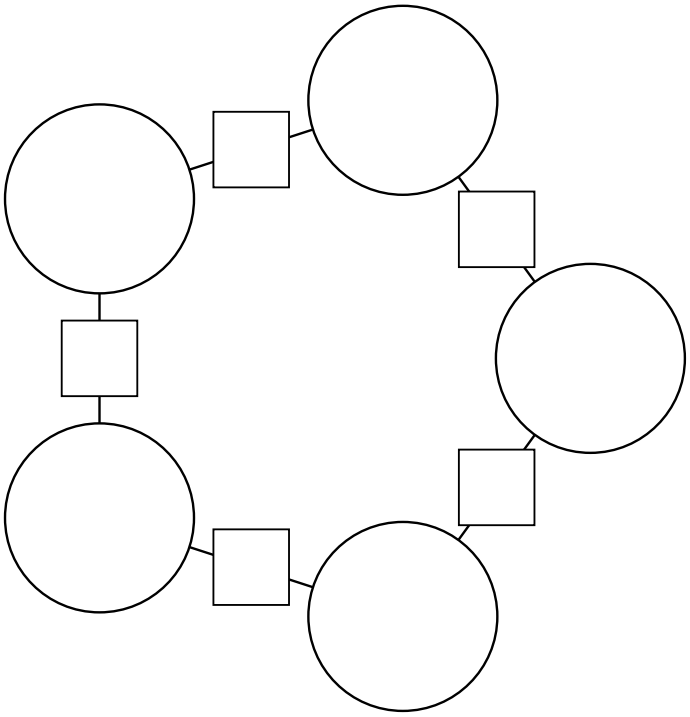
Problem 3: Label each loop, or cross it out if it cannot be done.



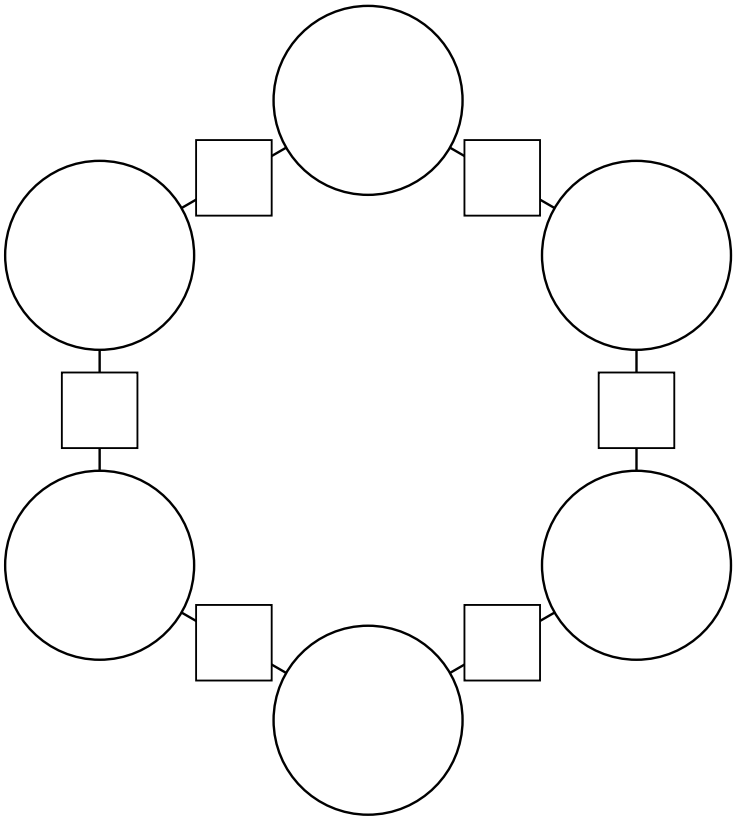
cards 0 to 3



cards 0 to 4



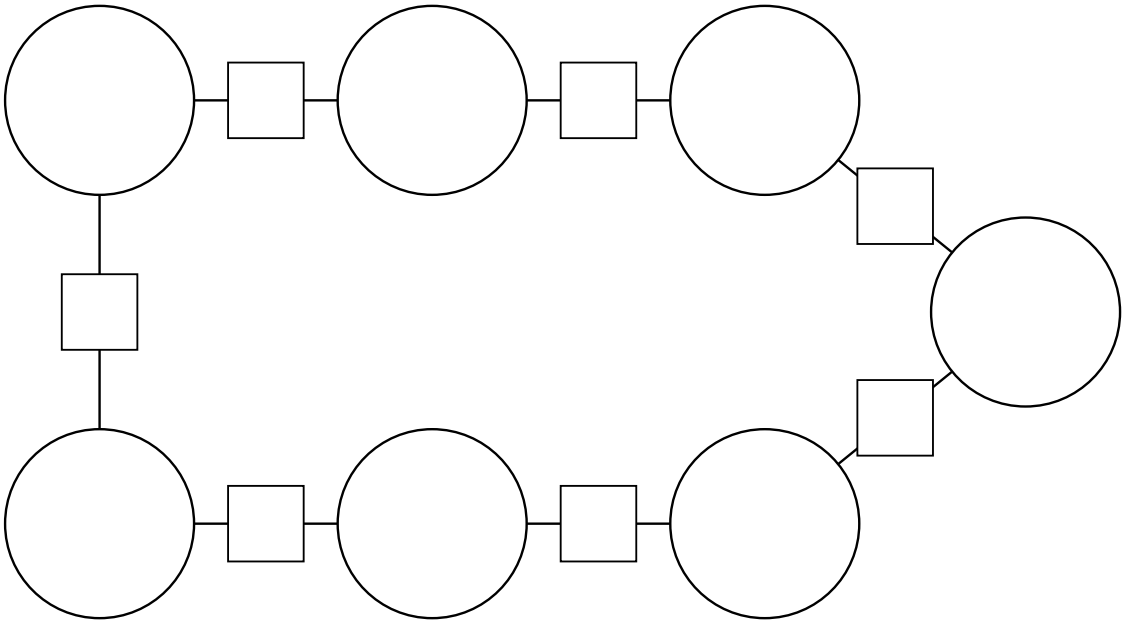
cards 0 to 5



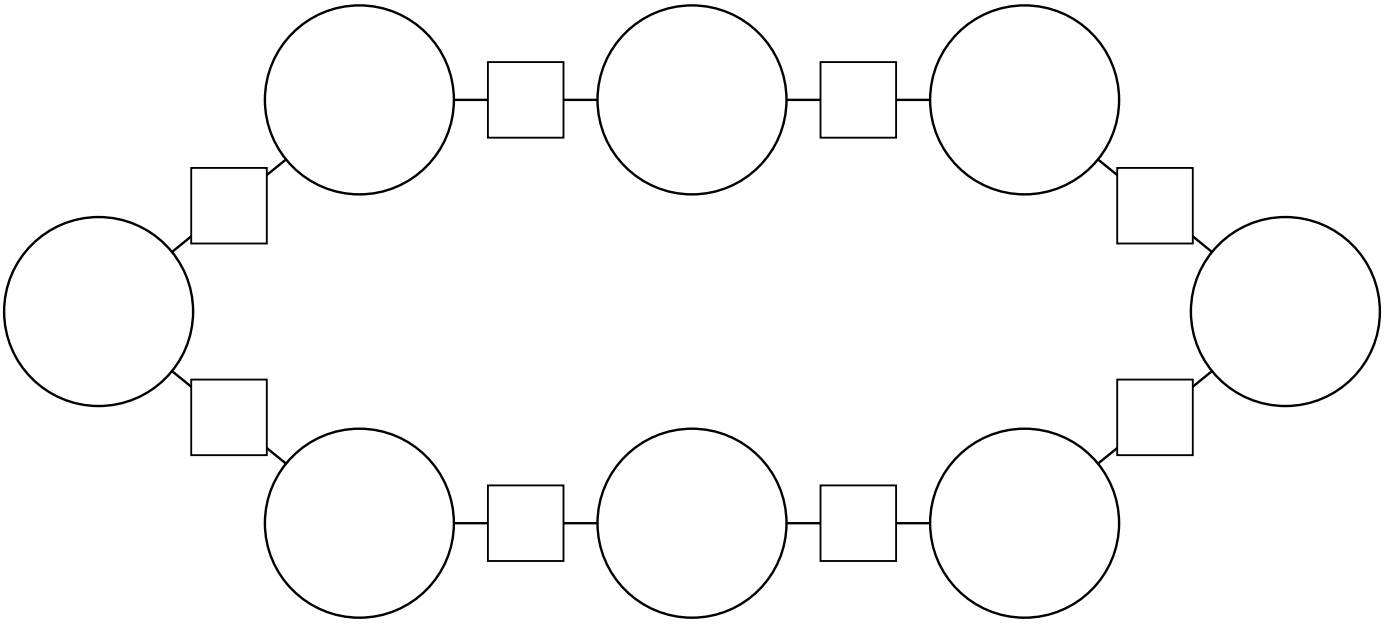
cards 0 to 6

Problem 4: Explain how you know that the loops you crossed out in Problem 3 cannot be labelled.

Problem 5: Label each loop, or explain why it cannot be done.



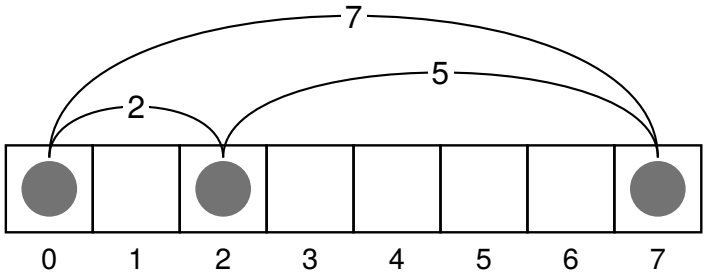
cards 0 to 7

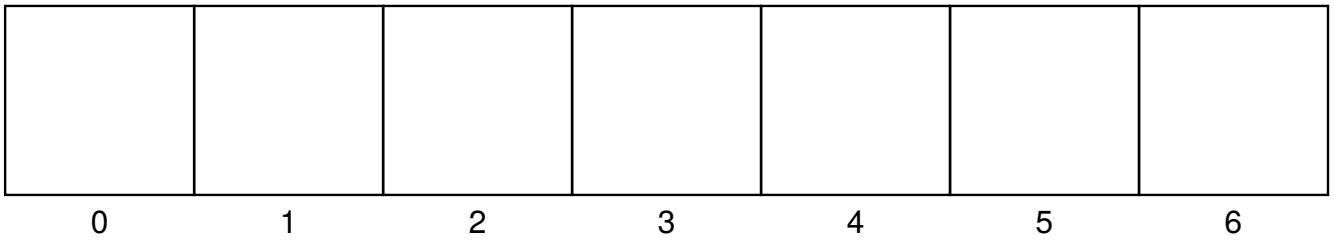


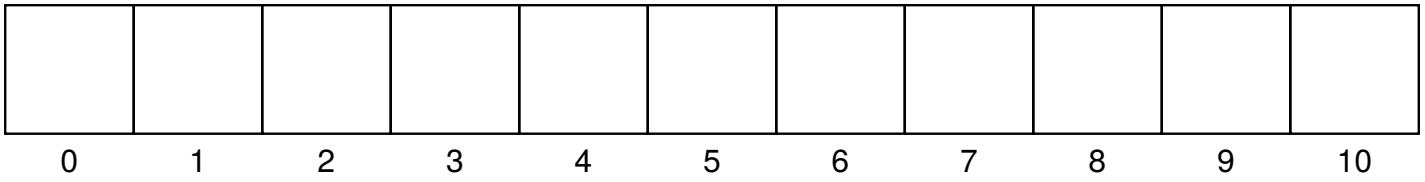
cards 0 to 8

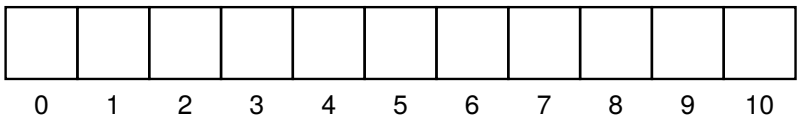
Problem 6: Which of these loops can you show cannot be labelled: 9, 10, 11, 12, 100 and 101 links?

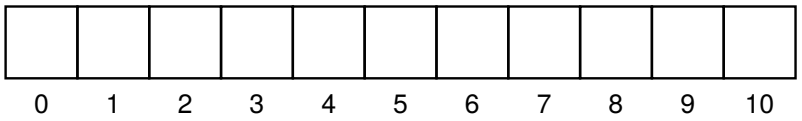
Problem 7: Put four counters on the strip from 0 to 6, one on 0 and one on 6, so that the six distances between pairs of counters are 1, 2, 3, 4, 5 and 6. Can you put five counters on the strip from 0 to 10, one on 0 and one on 10, so that the ten distances are 1 to 10?

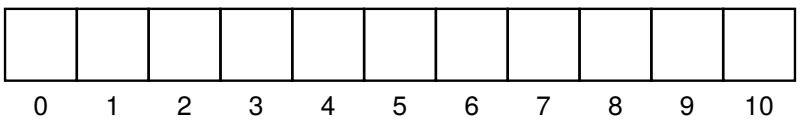




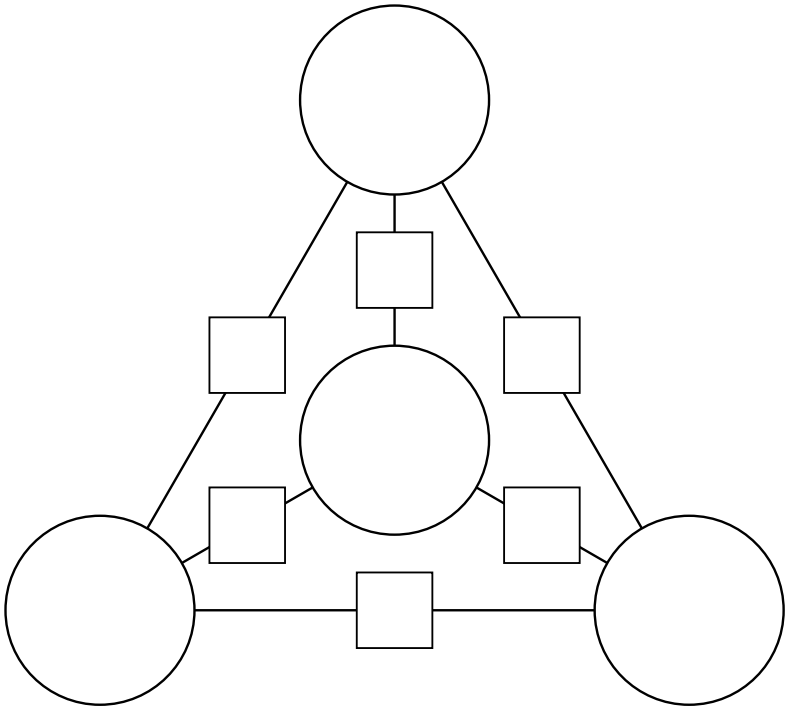




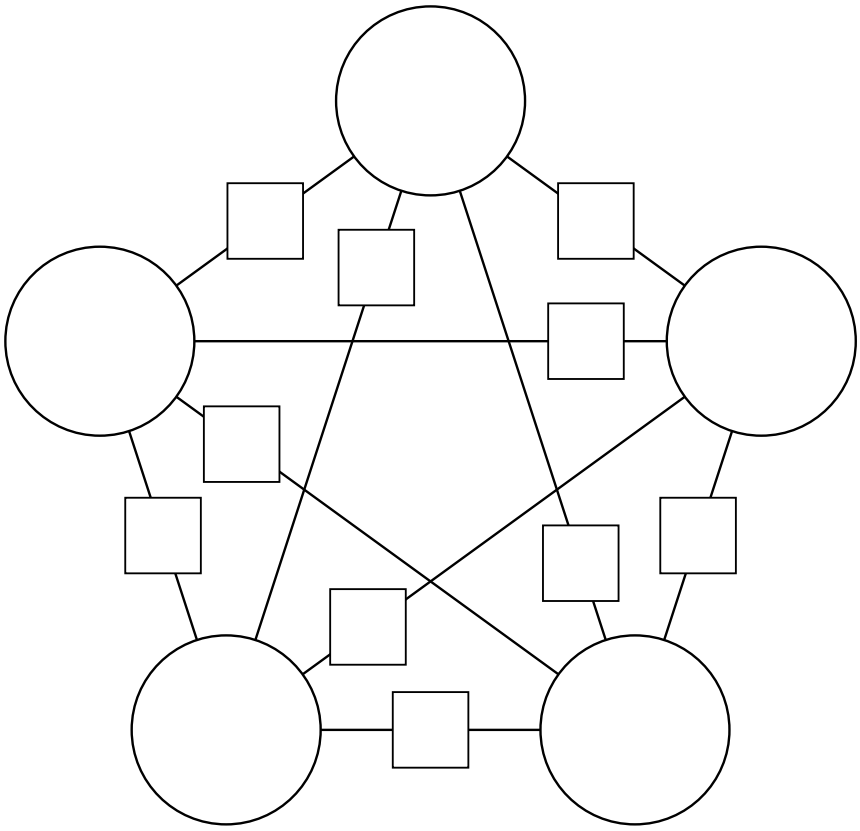




Problem 8: Label each network, or explain why it cannot be done.

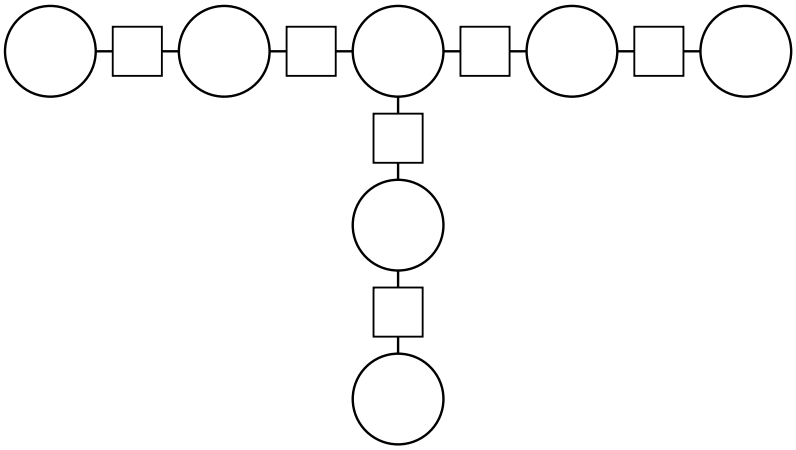


cards 0 to 6

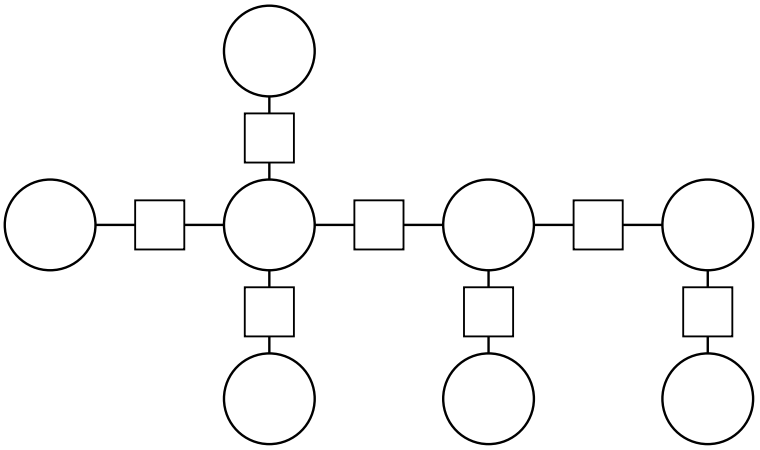


cards 0 to 10

Problem 9: A tree is a network in one piece with no loops. Label these trees. Then draw trees of your own with 7, 8 or 9 dots and label them. Can you find a tree that cannot be labelled?



cards 0 to 6



cards 0 to 7